Doubt session

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double x

int n → x^n

$2^{-2} = \frac{1}{2^2} = \frac{1}{4} = 0.25$

```
if(n<0){
    n=-n;
    x=1/x;
}
double pow=1;
while(n>0){
    pow=pow*x;
    n--;
}
Sout(pow);
```

→ n → x^n

→ x

n<0

$2^{-2} = \frac{1}{2^2}$

1) n → becomes positive.

2) $x \to \frac{1}{x}$

n → -ve

$-(-2) \to (+2)$

```
int m=n;
if(m<0){
    m=-m;
    x=1/x;
}
double Pow=1;
while(n>0){
    if(n%2!=0){
        Pow=Pow*x;
    }
    x=x*x
    n=n/2;
}
return Pow;
```

• -2,147,483,648 to 2,147,483,647

$m = -2147483648$

$m = 2147483648$

long m =

int m = 2147483648

Q. $(2)^{16} \to (2^2)^8 \to 2^{16}$ x=2, n=16

Q. $(4)^8$ x=4, n=8 → $(4^2)^4$ $(16^2)^2$

Q. $(16)^4$ x=16, n=4

Q. $(256)^2$ x=256, n=2

$(65536)^1$ x=65536 n=1 %2==0

→ return

1/2 ==0

→ x = x*x;
→ n = n/2

$(2)^5 = 2 \times 2^4 = 2^5$

$\frac{5}{2} = 2$

$2 \times (2^2)^2$

$2 \times (4)^2$

$\boxed{2 \times (4)^2}$

x = 2, n = 5

Pow = 1

Pow = 1*2

Pow = 2

x = 4 n = 2

x = 16 n = 1

n==1

Pow = Pow * x;

return Pow;

2^{15}

2×2^{14}

$2^{14} \to (2^2)^7$

Pow = 1*2

Pow = 2

$(4)^7$

Pow = 2*4*16

$4 \times (4)^6 \to (4^2)^3$

$(16)^3$

int m = 2147483648

long m =

$(16)^3$

$16 \times (16)^2$

256 n==1 (odd)

Pow = 2×4×16×256

return Pow;

Q. $(2)^{16}$ → even

$\hookrightarrow (2^2)^8 \rightarrow 2^{16}$

x=2, n=16

x = x×x

n = n/2

$(2^{16}) = (16)^4$

double Pow=1;

while( ){

if(n%2!=0) pow = Pow×x;

Q. $(4)^8 \rightarrow (4^2)^4$

x=4, n=8

x = x×x

Q. $(16)^4$

x=16, n=4

$(65536)^1$

int/ 1/2 = 0

Pow = 1×65536

Pow → Contains answer

Q. 2^5 → even

2×2^4 x=2, n=4

Pow = 1×2

$(2^2)^2 = 4^2$ x=4, n=2

Pow = 1×2×4

2^{25} → even

2×2^{24} n=24, x=2

Pow=1

$(2^2)^7 = 4^7$ → Odd

if (n%2!=0)

$\hookrightarrow$ Pow = Pow×x;

Pow = 2×4×16×256

$= 4 \times 4^6$

$(4^2)^3$

$(16)^3$

return ans;

Pow=1×2×4×16×256

Pow → Contains answer

$16 \times (16)^2$

x = 256 / n=1

Q. Write a function to return

Q. Write a function to return Square root of an int.

→ return value in int

$x = \boxed{256}$ / $\boxed{n=1}$ → NO

leetcode

```
public int Sqrt(int n){
    int i=1;
    while (i*i<=n){
        i++;
    }
    return i-1;
}
```

n=9

i=1*1 i=1

$\boxed{i=2}$ 2×2 c=9

i=3 3×3 =9 >8

n=4 → o/p =2

n=8 $\sqrt{8}$ → o/p = 2

n=15 → o/p=3

→ 3.....

3

Sqrt(n) $\sqrt{n}$

```
int ans=0;
for(int i=1; i<=n; i++){
    if(i*i > n) break;
    ans= i;
}
return ans;
```

i=1 $\boxed{i=2}$ i=3 ✗

$\boxed{ans=\cancel{1}\ 2}$

$\boxed{ans=i}$

---

n → palindrome → yes / NO

Rev the digits

Continue

```
int rev=0;
int temp=n;
if(n<0) return false;
while (temp!=0){
    int digit = temp%10;
    rev=rev*10+digit;
    temp= temp/10;
}
```

123 → 321

3→32

3×10+2

=32

32×10+1

=321

```
if(rev ==n) return true
return false;
```

```
}
if(rev==n) return true;
return false;
```